Chapter 10

Inter-VLAN Routing and DHCP

10.1 Inter-VLAN Routing

Let's start with the network shown in Fig. 10.1. In the network, Fa0/1 and Fa0/2 interfaces of SW1 belong to VLAN 10 and VLAN 20, respectively. After creating the VLANs and adding interfaces to them, when we ping from PC0 to PC1, the PC0 fails to communicate with PC1 as shown in Fig. 10.2. That is, an end device of one VLAN cannot communicate with an end-device of another VLAN. To enable the communication, we have to configure inter-VLAN routing. This can be done in several ways. However, the most popular technique is to use a router on a stick (ROAS) technique which will be discussed next.

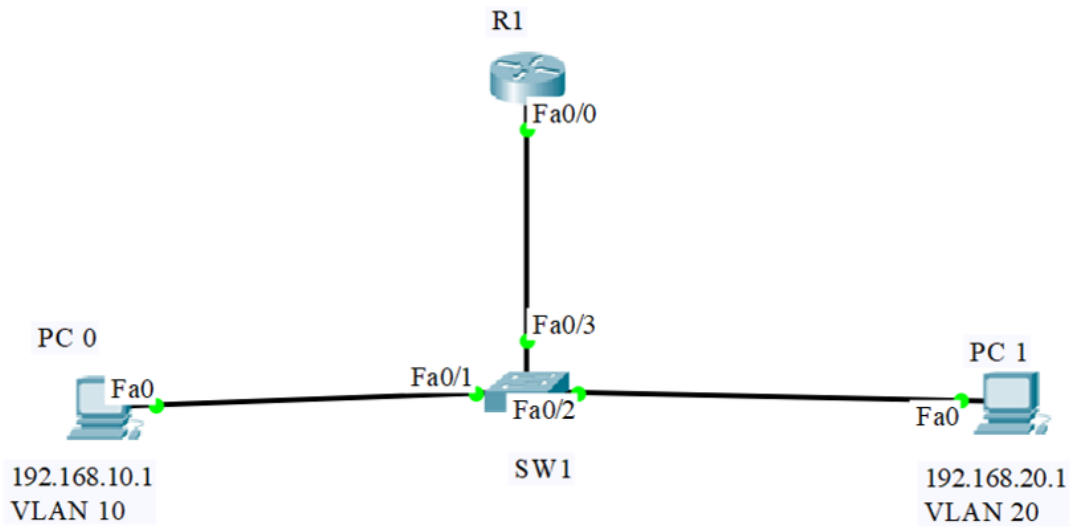


Fig. 10.1 A network with two VLANs and a ROAS.

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig. 10.2 Ping output before inter-VLAN routing.

In ROAS technique, the switch in which the VLAN is created and configured is connected to a router. The switch's interface connected to the ROAS is configured as a trunk. In Fig. 10.1, the link connecting SW1 and R1 is connected to the Fa0/3 interface of SW1; hence Fa0/3 has to be configured as a trunk. The IOS command is

```
Switch(config)#int fa0/3
Switch(config-if)#switchport mode trunk
```

Next, we configure the R1's Fa0/0 interface which the trunk line is connected to. First select the interface (line 1) and make it UP (line 2). Then we have to create sub-interface of the Fa0/0 interface. To create a sub-interface of an interface, we have to write interface Fa0/0.sub_interface_number. For example, we can write interface Fa0/0.1 or interface Fa0/0.2 and so on. The maximum number of sub-interfaces of an interface supported by a router depends on the model of the router. In line 3, we have created a sub-interface of the Fa0/0 interface and given the sub-interface number Fa0/0.10. When a trunk carries frames of multiple VLANs, the frame originating switch of the trunk adds a tag to each frame to indicate the VLAN the frame will be destined to. To indicate such information, some extra bits are added to the frame. How much extra bits will be added and what kind of information will be added depend on the protocol. Dot1q is a standard protocol for doing this. There is another protocol, named inter switch link (ISL) which is a cisco's proprietary protocol for doing the same task. However, the former one, dot1a, is widely used. Thus, after creating the sub-interface, we have to specify the protocol. In line 4, we have used dot1q protocol for VLAN 10. In other words, we can say that Fa0/0.10 sub-interface will be used for carrying data of VLAN 10 and dot1q protocol will be used for tagging purpose. Then, we assign and IP address to the sub-interface. This IP address must be valid host IP of the VLAN 10 and will be considered as the default-gateway of the VLAN 10. Thus, when we configure a PC which belongs to the VLAN 10, we have to input 192.168.10.254 (or any other valid host IP of the VLAN) as the default gateway.

Similarly, we can create another sub-interface for another VLAN; one sub-interface for each VLAN is needed. Please note that we have given the sub-interface name fa0/0.10 for the VLAN 10. However, it is not mandatory to use the VLAN's number in the sub-interface name. If you name the sub-interface as fa0/0.4 for the VLAN 10, there is nothing wrong with that. After configuring ROAS, we see in Fig. 10.3 that PC0 can communicate with PC1.

```
1.    Router(config)#interface fa0/0
2.    Router(config-if)#no shutdown

3.    Router(config-if)#interface  fa0/0.10
4.    Router(config-subif)#encapsulation dot1q 10
5.    Router(config-subif)#ip address 192.168.10.254 255.255.255.0

6.    Router(config-subif)#interface  fa0/0.20
7.    Router(config-subif)#encapsulation dot1q 20
```

8. `Router(config-subif)#ip address 192.168.20.254 255.255.255.0`

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=1ms TTL=127
Reply from 192.168.20.1: bytes=32 time<1ms TTL=127
Reply from 192.168.20.1: bytes=32 time<1ms TTL=127
Reply from 192.168.20.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig. 10.3 ROAS enables the communication between two VLANs

10.2 DHCP

Dynamic host configuration protocol (DHCP) is used for assigning IP addresses to hosts automatically by a DHCP server. A server can be configured to work as a DHCP server. As an alternative, a router can also be configured to work as a DHCP server. In this Lab, we will see how we can use a router as a DHCP server.

Let's start with the network shown in Fig. 10.4. There are two PCs, PC3 and PC4, which have not been assigned any IP address. The only one interface of the router is connected to the LAN. We are going to configure the router to use it as a DHCP server. Suppose that the network IP address is 192.168.1.0/24. In this block, there are 256 addresses out of which 192.168.1.1 to 192.168.1.254 are valid host IP address. First, configure the Fa0/0 interface of the router. Since this interface is connected to the LAN, we have to select any valid IP address from the above-mentioned range. Following the convention of default gateway IP address, we are selecting the last valid host IP address. Thus, the command of configuring Fa0/0 interface is given by

```
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shutdown
```

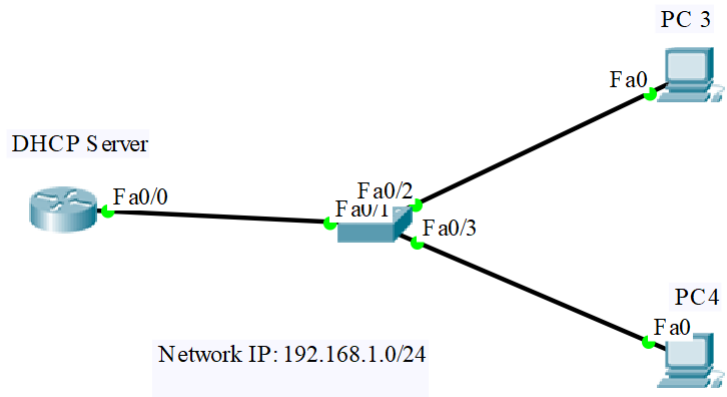
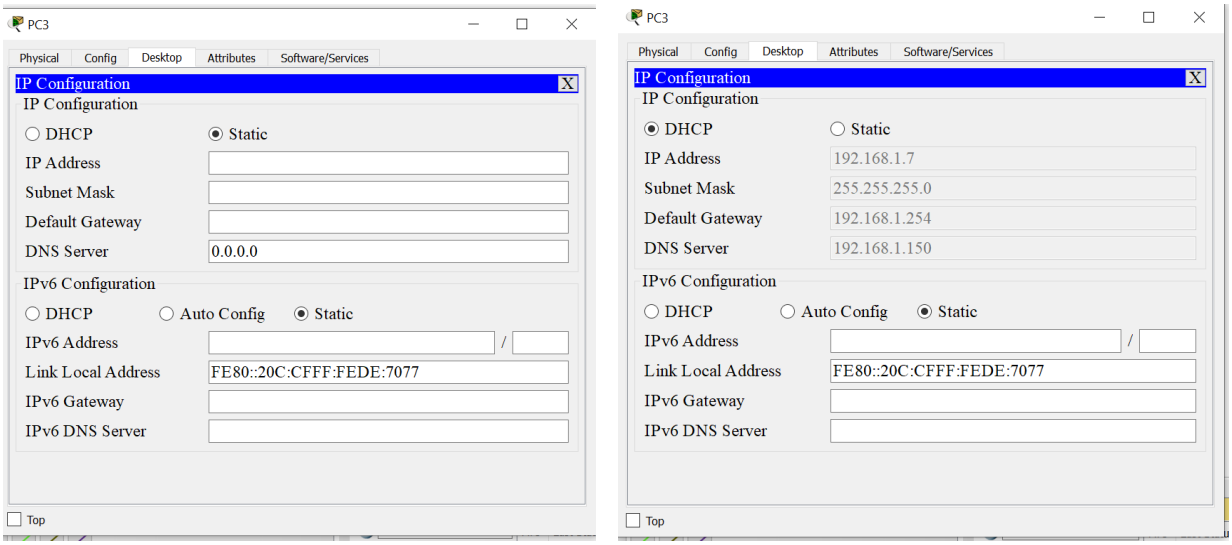


Fig. 10.4 Scenario: DHCP server in the same LAN

Now we will configure the router to make it a DHCP server. The commands are given below. In line 1, we first provide a command to create a pool (list of IP addresses to be allocated) of IP address from where we can allocate IP addresses to hosts. The line 1 creates the pool where the name of the pool is given student. We can give any other name. In line 2, the range of IP address is defined. The network IP address is given with its subnet mask. All the valid host IP addresses are included in the pool. In line 3, the default gateway is set to 192.168.1.254. When this DHCP server allocate an IP address to a host, it also sends the default gateway of this LAN. A DNS server is also set. Though we have not shown the DNS server in Fig. 10.4, we have used the command just to show how we can set a DNS server IP address in a DHCP server. Thus, in addition to the IP address, subnet mask, and default gateway (shown as default router), the DNS server address is also sent to an address seeking host. In the fifth line, we exclude few addresses from the pool. Although all valid host IP addresses are included in the pool in line 2, we can exclude a range of IP address from the pool by using the command shown in line 5. In this example, we have excluded 192.168.1.2 to 192.168.1.6, these six IP addresses. So, these three IP addresses will not be allocated to any host. To exclude multiple ranges of IP addresses, we have to use the command multiple times.

```
1.    Router(config)#ip dhcp pool student
2.    Router(dhcp-config)#network 192.168.1.0 255.255.255.0
3.    Router(dhcp-config)#default-router 192.168.1.254
4.    Router(dhcp-config)#dns-server 192.168.1.150
5.    Router(dhcp-config)#ip dhcp excluded-address 192.168.1.2
      192.168.1.6
```

Let's see the effect of the DHCP server configuration. In Fig. 10.5, we see PC3 has empty configuration when static configuration is selected because we have not done any configuration manually. However, when we select DHCP, the PC receives its IP address, subnet mask, default gateway (default-router) IP address and DNS server IP address from the DCHP server. Comparing the IP addresses allocated to PC3 and PC4 (see Fig. 10.5(b) and Fig. 10.6), we see that IP addresses 192.168.1.1 and 192.168.1.7 have been allocated and the addresses in between them have been excluded from the allotment due the 5th line of the commands.



(a) *(b)*

Fig. 10.5 PC3 configuration (a)Before DHCP (b) After DHCP

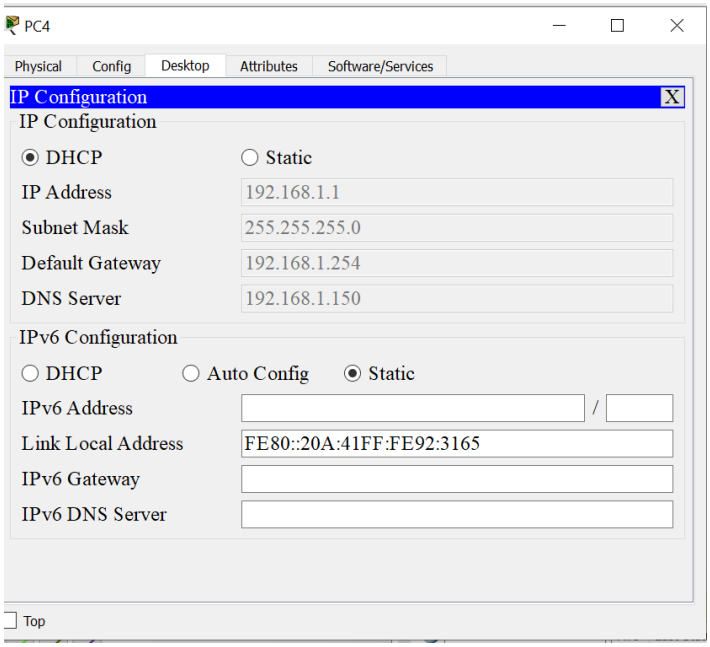


Fig.10.6 IP configuration of PC4

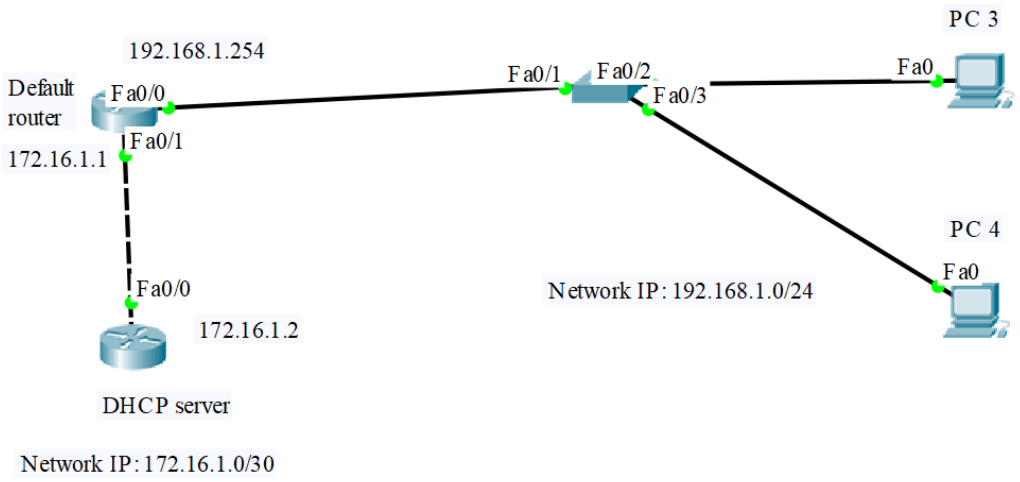


Fig.10.7 Scenario: DHCP server is in different network

Now, consider the network shown in Fig. 10.7 where the DCHP server is not located inside the LAN for which IP addresses have to be allocated automatically. In this case, when any PC broadcasts a DCHP discovery message with address 255.255.255.255. The packet will be blocked by the default router. Then, how is it possible for any host of the LAN to make its DHCP discovery message reachable to the DCHP server 172.16.1.2? To make it possible, configure the Fa0/0 and Fa0/1 interface of the default-router as follows:

```
Default-router(config)#int fa0/0
Default-router(config-if)#ip add 192.168.1.254 255.255.255.0
Default-router (config-if)#no shutdown

Default-router(config)#int fa0/1
Default-router (config-if)#ip add 172.16.1.1 255.255.255.252
Default-router (config-if)#no shutdown
```

Then configure the Fa0/0 interface of the DHCP server as

```
Router(config)#int fa0/0
Router(config-if)#ip add 172.16.1.2 255.255.255.252
Router(config-if)#no shutdown
```

Now, Configure the DHCP server as before

```
Router(config)#ip dhcp pool student
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.254
Router(dhcp-config)#dns-server 192.168.1.150
Router(dhcp-config)#ip dhcp excluded-address 192.168.1.2
192.168.1.6
```

Next task is to configure routing protocol to enable communication between two networks 192.168.1.0/24 and 172.16.1.0/30. We can do this using any routing protocols such RIPv2, EIGRP, or OSPF. This task is being left on the reader.

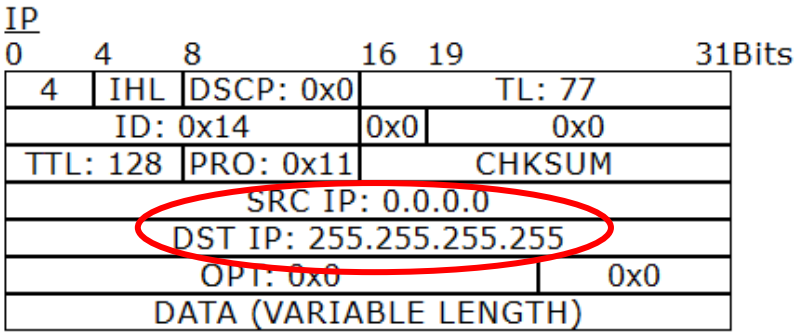
```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 172.16.1.0
Router(config-router)#no auto-summary
Router(config-router)#exit

Default-Router(config)#router rip
Default-Router(config-router)#version 2
Default-Router(config-router)#network 192.168.1.0
Default-Router(config-router)#network 172.16.1.0
Default-Router(config-router)#no auto-summary
Default-Router(config-router)#exit
```

After doing all these things, if we select DHCP in any PC of network 192.168.1.0/24, the DCHP request will fail because the DCHP discovery message will be blocked by the default router considering the message as usual broadcast message. To avoid this problem, we have to keep the default router informed to allow any DHCP message when it receives it. To do this, the default-gateway interface has to be configured by the following command as well:

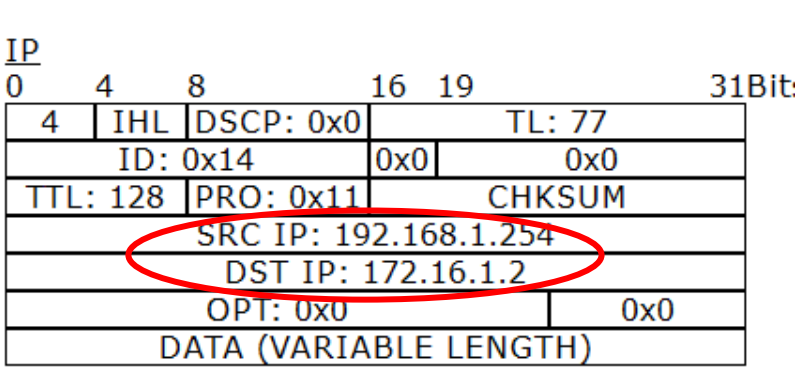
```
Default-Router(config-if)#int fa0/0
Default-Router(config-if)#ip helper-address 172.16.1.2
```

where 172.16.1.2 is the DHCP server address. Note that the previous IP configuration of the fa0/0 interface of the default router must be done.

When a packet travels from source to destination where the source and destination are located in different networks, the frames (Layer 2 protocol data unit (PDU)) gets changed in every hop (see Fig. 10.8). However, we see from the Layer 3 PDU (Packet) that the source and destination addresses are getting changed when a DHCP discovery message arrives at the default router from the source. In the incoming packet from PCX, the source IP is 0.0.0.0 while destination is 255.255.255.255 (Layer 3 broadcast IP). However, these two addresses get changed before the packet leaves the default router toward the DHCP server. The default gateway IP becomes the source IP while the DHCP server IP becomes the destination IP. After receiving the DHCP discover message, the DHCP server sends a DHCP offer message with its own address 172.16.1.2 as the source address while the default gateway interface of default router as the destination address. At the default router, the message's source address is changed to 192.168.1.254 and the message is broadcast to 255.255.255.255 (see Fig. 10.9).

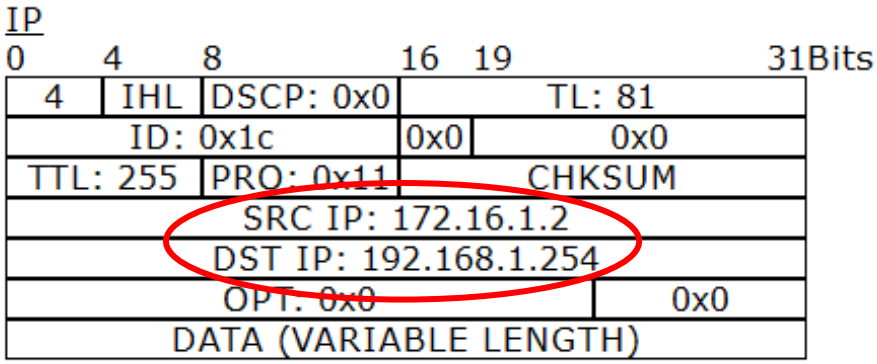


(a) Incoming packet header at default router from PCX



(b) Outgoing packet header from default router toward DHCP server

Fig. 10.8 Packet header of DHCP discover message at default router



(a) Incoming packet header at default Router from DHCP server

IP

Fig. 10.9 Packet header of DHCP offer message at default router